

SEC P vs NP log 2026-05-18

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-18 04:25 UTC

SEC P vs NP — daily log 2026-05-18

4 cycles recorded

Signed-Laplacian Holonomy Spectrum Bounds Tseitin Tree-Resolution

- **Verdict:** INCONCLUSIVE
- **Bridge:** Discrete magnetic / signed Laplacian spectral theory of finite graphs — the $\mathbb{Z}/2$ Bochner-Laplacian L^σ for a line bundle on G representing a cohomology class in $H^1(G; \mathbb{F}_2)$, accessed through the gauge-invariant smallest eigenvalue $\lambda_{\min}(L^\sigma)$ (Lieb-Loss 1993 on magnetic operators; Sunada 1994 on discrete magnetic Schrödinger; Mathai-Yates 2006; Atay-Liu 2014 and Lange-Liu-Peyerimhoff-Post 2015 on signed Cheeger inequalities). An arXiv search for ‘magnetic Laplacian’ AND (‘Tseitin’ OR ‘resolution refutation’ OR ‘proof complexity’) returns 0 direct hits and <5 adjacent papers, none using gauge-invariant signed spectra as a CNF invariant; distinct from sandpile / 2-Sylow / Ihara-zeta attempts on Tseitin (those use cokernels, non-backtracking matrices, or critical-group orders, never the $\mathbb{F}_2$ Bochner spectrum). $\times$ Tree-like Resolution / DPLL refutation size $t(T(G, \omega))$ for Tseitin XOR formulas $T(G, \omega)$ on connected 3-regular graphs G with odd charge ω (Urquhart 1987 / Ben-Sasson-Wigderson 2001 / Alekhnovich-Razborov 2001 expansion regime), in the Cook-Reckhow-Krajíček bounded-arithmetic framework where $\neg T(G, \omega)$ is Σ^b_0 and $\log_2 t$ controls V^0 -witnessing complexity.
- **Recorded:** 2026-05-18 00:33 UTC
- **Entry ID:** 2d264de23a0e

Statement

For a connected 3-regular graph $G=(V,E)$ with odd charge $\omega:V\rightarrow\mathbb{F}_2$ ($\sum_v \omega(v)\equiv 1 \pmod 2$), build a $\mathbb{Z}/2$ -gauge representative $\psi\in\mathbb{F}_2^E$ by BFS from a fixed root v_0 , setting each tree edge so that $\oplus_{\{e\ni v\}}\psi(e)=\omega(v)$ at every non-root v (the unsatisfiability deficit lands at v_0); form the signed adjacency A^σ with $\sigma(e)=(-1)^{\psi(e)}$ and the signed Laplacian $L^{\sigma=D-A^\sigma}$, and define $\mu(G,\omega):=\lambda_{\min}(L^\sigma)$, which is gauge-invariant under $\psi\mapsto\psi\oplus\delta f$ and equals 0 iff $\sum\omega$ is even. We conjecture an absolute constant $c>0$ such that for every unsatisfiable $T(G,\omega)$ on 3-regular G , $\log_2 t(T(G,\omega))\geq c\cdot\mu(G,\omega)\cdot|V|$, with pre-registered $c=0.05$. A single 3-regular instance (G,ω) with $\log_2 t(T(G,\omega))<0.05\cdot\mu(G,\omega)\cdot|V|$ falsifies the conjecture.

Rationale

The Tseitin obstruction is exactly the $\mathbb{Z}/2$ holonomy class $[\omega]\in H^1(G;\mathbb{F}_2)$, and $\mu(G,\omega)$ is the spectral norm of this obstruction in the discrete Bochner-Laplacian sense — large ($\Theta(1)$) on expanders where holonomy is globally rigid, and small ($\Theta(1/|V|^2)$) on tree-like graphs where the obstruction concentrates locally. The mechanism funnels the Ben-Sasson-Wigderson expansion-based $\exp(\Omega(\text{boundary expansion}))$ lower bound through a single gauge-invariant scalar via the signed Cheeger inequality $h^\sigma(G)\geq\sqrt{(2d\mu)}$, making the standard expansion bound a corollary of a finer structural invariant.

The invariant is a property of the CNF instance (not of any truth-table), so it is shielded from natural proofs; it does not algebrize (signed Laplacian eigenvalues are not preserved under polynomial extension of the Boolean ring); and Tseitin lower bounds for tree-Resolution are not relativizing-style oracle facts.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.12s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_92felba4.py", line 158, in <module>
    result = run_trial(seed)
            ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_92felba4.py", line 126, in run_trial
    psi = bfs_gauge(vertices, edges, 0)
         ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_92felba4.py", line 111, in bfs_gauge
    if adj[u][v] == 1 and not visited[v]:
       ^^^
```

NameError: name 'adj' is not defined

Judge reasoning

Test crashed due to undefined variable 'adj' in BFS gauge calculation, preventing evaluation of conjecture validity | next: Fix the 'adj' variable definition in bfs_gauge function and rerun tests with multiple seeds to assess support fraction

Bonami-Beckner 4-2 Saturation of Clause-Sum Bounds DPLL Depth

- **Verdict:** INCONCLUSIVE
- **Bridge:** Bonami-Beckner hypercontractivity (Bonami 1970, Beckner 1975) — the degree-d 4-norm-vs-2-norm contraction $\|P\|_4 \leq (q-1)^{\{d/2\}} \|P\|_2$ — applied as a SYNTACTIC Fourier-analytic invariant of the centered clause-indicator polynomial Q_F of a CNF (a property of the formula's clause/polarity structure, NOT of its truth table). An arXiv search 'Bonami-Beckner hypercontractivity' AND ('DPLL' OR 'tree-Resolution' OR 'CNF refutation') returns 0 direct hits, with <5 adjacent papers (Schoenebeck-style PC-degree, never DPLL search-tree depth via syntactic clause-sum 4-2 ratios). Distinct from the blacklisted Cross-Side Fourier attempt: that invariant was computed on the truth-table communication MATRIX (natural-proof shape); here Q_F lives on the clause-polarity STRUCTURE of F , so two semantically equivalent CNFs F_1, F_2 can have very different $r(F)$, shielding the invariant from Razborov-Rudich constructivity. $\times$ DPLL / tree-like Resolution refutation depth $d(F)$ for unsatisfiable 3-CNFs, in the Ben-Sasson-Wigderson-Krajíček bounded-arithmetic regime where $\log_2 d(F)$ controls V^0 -witnessing complexity of $\neg F$ as Σ^b_0 and lower-bounds tree-Resolution width $w^*(F)$.
- **Recorded:** 2026-05-18 03:29 UTC
- **Entry ID:** 2b2653395609

Statement

For an unsatisfiable 3-CNF F with m clauses on n variables, let $s_{\{i,v\}} \in \{\pm 1\}$ be the polarity sign of variable v in clause C_i ($s=+1$ if v occurs negated, -1 if positive). Define the centered

clause-falsification polynomial $Q_F: \{-1, +1\}^n \rightarrow \mathbb{R}$ by $Q_F(x) = \sum_{i=1}^m (\prod_{v \in \text{vars}(C_i)} (1 + s_{i,v}x_v)/2 - 1/8)$, whose Fourier coefficient at $T \subseteq [n]$ with $|T| \in \{1, 2, 3\}$ is the closed-form sum $\hat{Q}_F(T) = (1/8) \sum_{i: T \subseteq \text{vars}(C_i)} \prod_{v \in T} s_{i,v}$ (computable directly from clause structure in $O(m)$ time). Let $r(F) := \|\hat{Q}_F\|_4 / \|\hat{Q}_F\|_2 \in [1, 3^{3/2}]$ be the Bonami-Beckner 4-2 ratio, and set $G(F) := (3^{3/2} - r(F)) / 3^{3/2} \in [0, 1 - 3^{-3/2}]$. Conjecture: there is an absolute constant $c \geq 0.05$ such that for every unsatisfiable 3-CNF F , $\log_2(d^*(F) + 1) \geq c \cdot n \cdot G(F)$; a single F violating this inequality falsifies.

Rationale

Hypercontractive saturation $r(F) \rightarrow 1$ means Q_F has a heavy-tailed degree-3 spectrum: clause-violation behaves like a sharp witness function whose Fourier mass spreads thin, the regime characterizing expander-Tseitin and threshold-random 3-SAT where DPLL must explore exponentially many literal interactions. Conversely $r(F) \rightarrow 3^{3/2}$ (Khintchine-saturated) means Q_F is dominated by a few large coefficients, the regime of structurally simple unsat formulas that DPLL refutes shallowly. The bridge is novel because the Fourier analysis is performed on the CLAUSE STRUCTURE polynomial Q_F (syntactic), not on F 's truth table or its communication matrix, sidestepping the natural-proofs largeness barrier that killed the Cross-Side Fourier attempt.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.18s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_df7c90d8.py", line 178, in <module>
    result = run_trial(seed)
             ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_df7c90d8.py", line 141, in run_trial
    norm_4 = compute_norm_4(F, n)
             ~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_df7c90d8.py", line 81, in compute_norm_4
    product *= (1 + s * x[v - 1]) / 2
                ~~~~~
```

IndexError: list index out of range

Judge reasoning

The test crashed with an IndexError in compute_norm_4 before producing any data, so the pre-registered support condition across 480 instances could not be evaluated. | next: Fix the variable indexing bug in compute_norm_4 (likely a 1-based vs 0-based mismatch or variables exceeding n) and rerun the full 480-instance sweep.

Cycle-Type Mixing Defect of Barrington BP Bounds NC^1 Length

- **Verdict:** INCONCLUSIVE
- **Bridge:** Character-theoretic mixing on S_5 — the L^1 distance between the empirical cycle-type distribution of partial products of a deterministic word and the Haar cycle-type distribution $\pi^* = (1, 10, 15, 20, 20, 24, 30)/120$, in the Diaconis-Shahshahani 1981 / Lulov-Pak 2002 / Larsen-Shalev 2008 Plancherel-spectrum tradition. This is a STRUCTURAL invariant of the program (a property of the BP wiring, not of its truth table), and an arXiv search ‘conjugacy

class mixing' AND ('branching program' OR 'Barrington' OR 'formula size') returns 0 direct hits with <5 adjacent papers, all on quantum-circuit Haar approximation. $\times NC^1$ formula leaf size via Barrington 1989 width-5 permutation branching programs over S_5 : every depth- d Boolean formula compiles into a length- 4^d width-5 BP over S_5 whose terminal permutation is the identity iff $f(x)=0$ and the fixed 5-cycle σ_α iff $f(x)=1$. The target separation is NC^1 vs explicit P functions; the structural form of the invariant shields it from the Razborov-Rudich natural-proofs barrier (two formulas with identical truth tables can compile to BPs with very different MIX).

- **Recorded:** 2026-05-18 03:58 UTC
- **Entry ID:** 816a64f0e4e2

Statement

Let $P=(I_1, \dots, I_L)$ be a width-5 BP over S_5 computing a Boolean f via Barrington's protocol, with $I_t=(v_t, \pi_t^{0, \pi_t^{-1}})$. For uniform $X \in \{0,1\}^n$ let $\mu_t^X \in P$ be the distribution on the 7 conjugacy classes of S_5 induced by $\Pi_{\{s \leq t\}} \pi_s^{X_{\{v_s\}}}$, and define $MIX(P) := (1/L) \cdot \sum_{t=1}^L \|\mu_t^{P^{-n}}\|_1$. Conjecture: for every non-constant f and every width-5 Barrington BP P computing f via the standard Ben-Or-Cleve commutator gadget, $MIX(P) \cdot L \geq (1/8) \cdot \log_2 \text{LeafSize}(f)$, where $\text{LeafSize}(f)$ is the minimum literal-leaf count of any AND/OR/NOT formula computing f . A single (P, f) with $MIX(P) \cdot L < (1/8) \cdot \log_2 \text{LeafSize}(f)$ refutes the conjecture.

Rationale

Barrington-compiled BPs realize commutators of 5-cycles whose conjugacy-class trajectory is forced to spend most steps in non-trivial classes when the underlying formula has high algebraic depth; π^* assigns 119/120 mass off the identity class, so the deterministic word can only stay 'near uniform in cycle type' by genuinely commutator-rich computation. Linking $MIX \cdot L$ to $\log \text{LeafSize}$ makes mixing a syntactic complexity certificate that is bypassed by no truth-table proxy, addressing the ACC_SIPSER target of finding natural properties that are STRUCTURAL (not truth-table). The bridge to Lulov-Pak character bounds promises a path to $ACC^0[m]$ via Barrington-Thérien compilation into solvable iterated wreath products without invoking arithmetization (avoiding ALGEBRIZATION).

Novelty

- Judge: NOVEL over 8 arXiv hits

Top hits consulted: - [1709.08890v2] Partial matching width and its application to lower bounds for branching programs - [1107.1826v4] Conjugacy growth of finitely generated groups - [1007.1875v2] Lower Bounds for Quantum Oblivious Transfer - [1512.02171v5] Right-jumps and pattern avoiding permutations - [1301.0379v2] The Parameterized Complexity of some Permutation Group Problems

Empirical Test

- exit code: 1, elapsed: 0.12s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c02f54b6.py", line 108, in <module>
    result = run_trial(seed)
    ~~~~~^~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c02f54b6.py", line 85, in run_trial
    metric_value = barrington_protocol(formula, n)
    ~~~~~^~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c02f54b6.py", line 52, in barrington_protocol
    product *= barrington_protocol(formula[bit], n)
    ~~~~~^~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c02f54b6.py", line 47, in barrington_pr
    if formula[bit] == 'x':
        ~~~~~^~~~~^
```

IndexError: string index out of range

Judge reasoning

The test crashed with an `IndexError` in `barrington_protocol` before producing any `RESULT` line or data, so neither the support nor falsification condition could be evaluated. | next: Fix the recursive formula parser (handle empty subformulas and properly tokenize literals/operators) and re-run the 30-seed sweep.

NW-Design Restriction Dimension Separates Perm from Padded Det

- **Verdict:** INCONCLUSIVE
- **Bridge:** Nisan-Wigderson combinatorial designs (set systems with bounded pairwise intersection from PRG construction; Nisan-Wigderson 1994; Impagliazzo-Wigderson 1997) $\times$ Mulmuley-Sohoni GCT determinantal complexity $dc(\text{perm}_n)$: orbit-closure separation of perm_n from padded det_m for $m \leq n^{\{1.5\}}$ (Mignon-Ressayre 2004; Landsberg 2017)
- **Recorded:** 2026-05-18 04:25 UTC
- **Entry ID:** 0cac01ce9ab0

Statement

Let $D=(S_1, \dots, S_M)$ be a Nisan-Wigderson design on the index set $[n] \times [n]$ of matrix-entry variables $x_{\{ij\}}$, with each $|S_t|=n$ and pairwise intersections $|S_s \cap S_t| \leq k = \lceil \log_2 n \rceil$. For a homogeneous degree- n polynomial $f \in \mathbb{C}[x_{\{ij\}}]^n$, define the NW-restriction dimension $\rho_D(f) := \dim_{\mathbb{C}} \text{span}\{f^{\wedge\{t\}}(t) : t \in [M]\}$, where $f^{\wedge\{t\}}(t)$ is the polynomial obtained from f by setting $x_{\{ij\}} \mapsto 0$ for every $(i,j) \notin S_t$. CONJECTURE: there exist absolute constants $C_1, C_2 > 0$ with $C_2 > 4C_1 \cdot \log_2 n / 1$ such that for every n and every M with $n \leq M \leq \lfloor n^{\{1.5\}} \rfloor$: (a) for every padded determinant $g(x) = \sum (x)^{\wedge\{n-m\}} \cdot \text{det}_m(L(x))$ with $m \leq \lfloor n^{\{1.5\}} \rfloor$, $\square$ a linear form and L an $m \times m$ matrix of linear forms in the $x_{\{ij\}}$, $\rho_D(g) \leq C_1 \cdot m \cdot \log_2 n$; (b) $\rho_D(\text{perm}_n) \geq C_2 \cdot M$. A single (n, M, D, g) violating (a) or a single (n, M, D) violating (b) refutes the conjecture.

Rationale

NW designs probe disjoint-up-to-log-overlap windows of variables, so the M restrictions act like ‘approximately independent samples’ of the polynomial’s syntactic content. A padded determinant compresses through only $O(m^2)$ hidden linear forms, so its NW-restrictions are forced to live in an $O(m \cdot \text{polylog})$ dimensional subspace; the permanent has no such linear-form bottleneck, and its restrictions should be near-linearly-independent across NW slots. Because ρ_D is an $\exp(n)$ -time syntactic invariant of the polynomial expansion (not of a truth table), it is shielded from Razborov-Rudich natural proofs, and the NW-design layer ties hardness-vs-randomness combinatorics to a GCT obstruction shape distinct from prior Specht/Mahonian/Eulerian attempts.

Novelty

- Judge: NOVEL over 3 arXiv hits

Top hits consulted: - [2202.13016v1] Bounds on Determinantal Complexity of Two Types of Generalized Permanents - [1007.3804v4] Symmetric Determinantal Representation of Formulas and Weakly Skew Circuits - [0106076v3] A determinantal formula for the Hilbert series of one-sided ladder determinantal rings

Empirical Test

- exit code: 1, elapsed: 0.10s

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b4a3a7c2.py", line 163, in <module>
    result = run_trial(seed)
            ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b4a3a7c2.py", line 130, in run_trial
    perm_rank = matrix_rank(perm_matrix)
                ^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b4a3a7c2.py", line 33, in matrix_rank
    while pivot < n and matrix[pivot][col] == 0:
            ~~~~~^~~~~
```

IndexError: list index out of range

Judge reasoning

The test crashed with an `IndexError` in `matrix_rank` before any trials completed, so no data was produced to evaluate the pre-registered support or falsification conditions. | next: Fix the `matrix_rank` indexing bug (likely a row/column dimension mismatch when building `perm_matrix`) and rerun the full 270-trial sweep.